Physics Assignment-1

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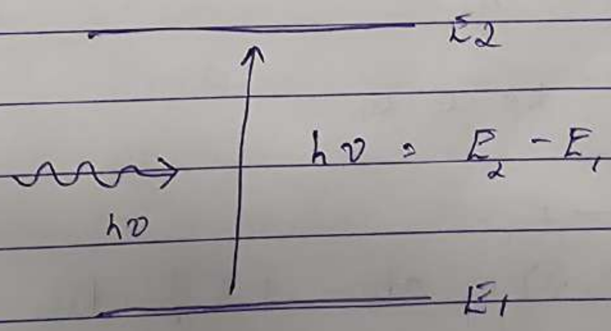
1) What is induced ~~included~~ absorption, spontaneous emission and stimulated emission? Explain with suitable diagram.

Absorption:

It is the process in which an atom absorbs energy from incoming radiation and an electron jumps from a lower energy level to a higher energy level.

Atom initially in ground state (E_1)
Photon with energy $h\nu = E_2 - E_1$ hits the atom

Electron moves to excited state (E_2)

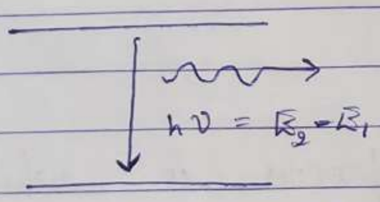


atom + photon ($h\nu$) = atom*
atom* represents an atom in excited state

→ Spontaneous Emission

It is the process in which an excited atom returns to a lower energy state on its own and emits a photon.

- Electron in excited state (E_2)
- After some time, it falls back to E_1
- Emits photon of energy $h\nu$



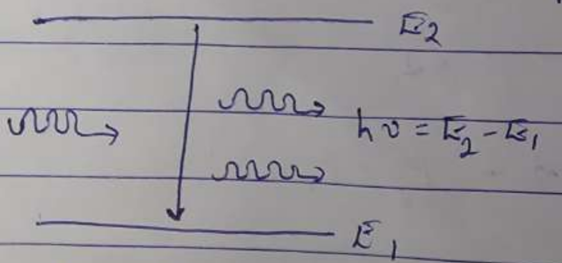
atom = atom + photon ($h\nu$)

The emitted photons have random nature (Incoherent).

→ Stimulated Emission

It Occurs when an incoming photon forces an excited atom to emit another photon.

- Atom already in excited state (E_2)
- Incoming photon ($h\nu$) interacts
- Atom emits a second photon and falls to E_1



atom* + photon ($h\nu$) = atom + 2 photons ($h\nu$)

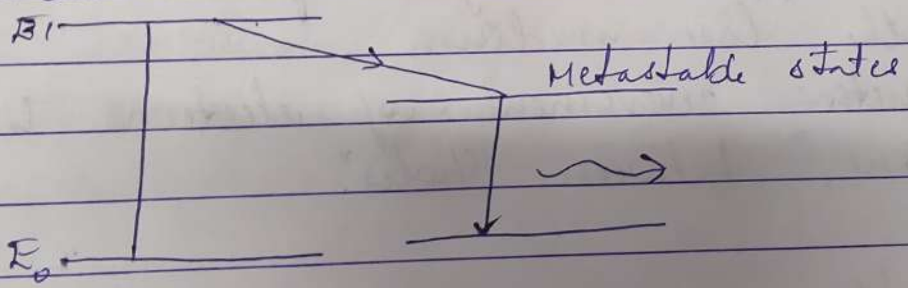
These two photons (Stimulating photon and stimulated photons) emerging out are in the same direction, phase, energy and frequency

Hence they are 'Coherent Photons'

2) Explain the role of each parameter in LASER emission

i) Metastable state

In the excited state, atoms stay only for $10^{-8}s$.
Metastable states are intermediate state, where the lifetime of atoms is $\approx 10^{-3}s$.
This property will help to achieve population inversion.



ii) Stimulated emission

i) Population inversion is created (more atoms in excited state than ground state)

ii) A photon of energy $h\nu$ interacts with an excited atom

iii) The atom emits a second photon due to stimulated emission

iv) Both photons are:
- Identical
- Travel in the same direction
- in phase (coherent)

v) These photons stimulate more atoms
→ chain reaction

vi) This leads to amplication of light, producing a strong laser beam

iii) Pumping
Supplying energy to the medium to excite atoms from lower energy state.

- Pumping can be done in diff. ways:
- Optical: flashlamps and high-energy light sources
- Electrical: application of a potential difference across the laser medium
- Semiconductor: movement of electrons in "junctions," between "holes"

iv) Active medium
It is a system or substance which has suitable energy level system so that population inversion and consequently light amplification is possible.

Optical cavity / Laser cavity:

It is the space between two mirrors or reflecting surface. Here, the photons travel repeatedly to and fro until a strong laser beam is obtained. The length of optical cavity is chosen in such a way that $L = m\lambda/2$. where, λ is wavelength of the laser and m is an integer.

Laser cavity

Mirror | Active medium | Mirror

$\longleftarrow L = m\lambda/2 \longrightarrow$

5) The wavelength of semiconductor laser diode is 800 nm, find the bandgap of semiconductor laser

⇒ Given:

$\lambda = 800nm = 8 \times 10^{-9} m$

$h = 6.626 \times 10^{-34} Js$

$c = 3 \times 10^{8} m/s$

$Eg = \frac{hc}{\lambda}$

$= \frac{6.626 \times 10^{-34} \times 3 \times 10^{8}}{3 \times 10^{8}}$

$= \frac{19.878 \times 10^{-26}}{800}$

$= 2.48 \times 10^{-19} J$

$1eV = 1.6 \times 10^{-19} J$

$E_g = \frac{2.48 \times 10^{-19}}{1.6 \times 10^{-19}}$

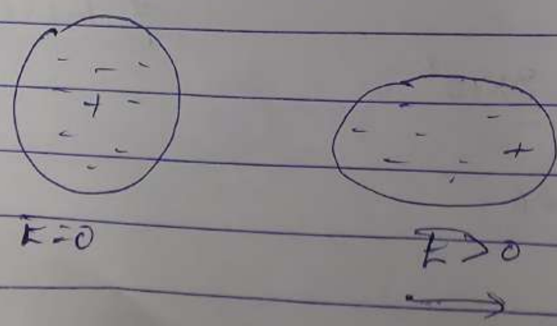
$= \boxed{1.55eV}$

4) Define polarisation. Explain types of polarization in detail with diagram.

The displacement of charges (or dipole moment) within an atom of a dielectric under an applied electric field is called polarisation.
The applied field induces polarisation, hence the induced dipole moment per unit volume of the dielectric is also called polarisation.
It is a vector quantity.

Types of Polarisation:

1) Electrical polarisation:- The electronic polarisation occurs due to the displacement of the positive charege and negative charges in the atoms of the dielectric material on the application of the external field.
This kind of polarisation is present in all materials. It is proportional to the volume of the atoms in the material and is independent of temperature.



Therefore, the electronic Polarisability α_e is proportional to the volume of the atom. Let N be the no of atoms in a unit volume of the dielectric then

dipole moment / unit volume = N x atomic dipole moment

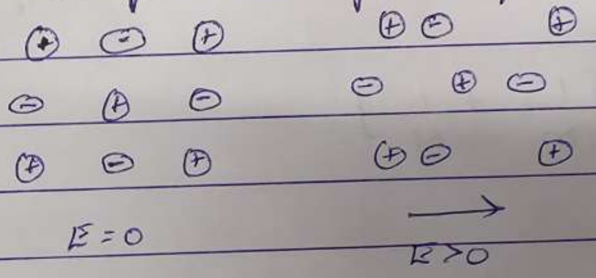
$$= N\alpha_e E$$

But dipole moment / unit volume $= P = \epsilon_0(\epsilon_r - 1)E$

$$\therefore N\alpha_e E = \epsilon_0(\epsilon_r - 1)E$$

$$\alpha_e = \epsilon_0(\epsilon_r - 1)/N$$

2) Ionic polarisation: Ionic polarisation occurs only in those dielectric materials which posses ionic bonds, such as NaCl and KCl. The ionic polarisation is due to the displacement of cations and anions in opposite directions when subjected to an external electric field in an ionic solid. This displacement is also independent of temperature.



Suppose an electric field is applied in the positive x direction. Then the positive ions move to the right and negative ions move to the left. We assume that there are one cation and one anion in each unit cell of that ionic crystal.

Due to ionic displacement, the resultant dipole moment per unit cell $\mu = e(x_1 + x_2)$ where
x_1 = Shift of +ve ion w.r.t equilibrium position
x_2 = shift of -ve ion w.r.t equilibrium position

Due to the application of static electric field, the force produced may be taken as F newton, and the restoring force on the +ve ion be $\beta_1 x_1$ and restoring force on -ve ion be $\beta_2 x_2$ where β_1 and β_2 are restoring force constants which depend upon the mass of the ion and the angular frequency of the molecule.

Under equilibrium $F = \beta_1 x_1 = \beta_2 x_2$

$x_1 = \frac{F}{\beta_1} = \frac{eE}{m\omega_0^2}$; m is the mass of +ve ion

$x_2 = \frac{F}{\beta_2} = \frac{eE}{M\omega_0^2}$; M is the mass of -ve ion

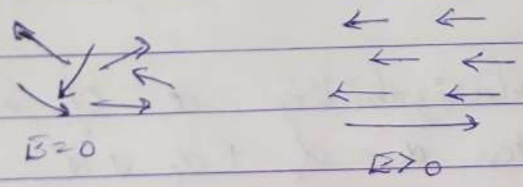
$x_1 + x_2 = \frac{eE}{\omega_0^2}\left[\frac{1}{m} + \frac{1}{M}\right]$

induced dipole moment $= \mu = e(x_1 + x_2) = \frac{e^2 E}{\omega_0^2}\left[\frac{1}{m} + \frac{1}{M}\right]$

Ionic polarisability $\alpha_i = \frac{\mu}{E} = \frac{e^2}{\omega_0^2}\left[\frac{1}{m} + \frac{1}{M}\right]$

3. Orientational polarisation:

Orientational polarisation occurs in polar dielectrics (dielectrics with permanent dipoles). The dipole orientation of these molecules is random due to thermal agitation hence net polarisation is zero. But under the influence of an applied electric field, each of the dipoles undergoes rotation so as to reorient along the direction of the field and the material develops electrical polarisation.

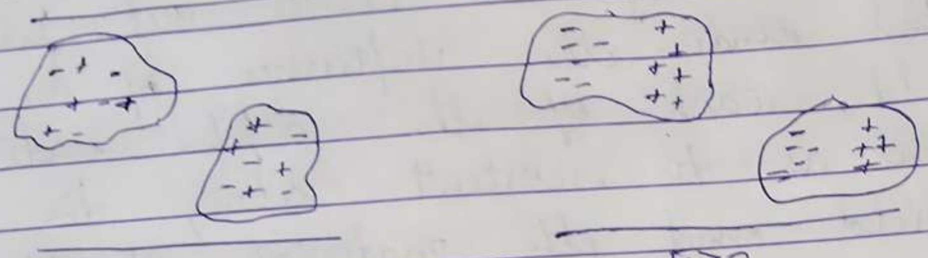


The orientation polarization is strongly temperature dependent and decreases with an increase in temperature. In the case of polar dielectrics, the oriental polarizability α_0 is given by

$$\alpha_0 = \frac{\mu^2}{3KT}$$

4) Space charge polarisation: Space charge polarisation occurs in multiphase dielectric materials in which there is a change of resistivity between different phases when such materials are subjected to an electric field, at high temperatures, the charges get accumulated at the interface ~~between~~ because of sudden change in conductivity across the boundary. Since the accumulation of charge

with opposite polarity occurs at opposite parts in the low resistivity phase. it leads to the development of a dipole moment within the low resistivity phase domain



Thus the total polarizability α of a material can be written $\alpha = \alpha_e + \alpha_i + \alpha_o$

$$= 4\pi\epsilon_0 R^3 + \frac{e^2}{\omega_0^2}\left[\frac{1}{m} + \frac{1}{M}\right] + \frac{\mu^2}{3KT}$$

Total polarisation $P = N\alpha E = NE$

$$\left\{4\pi\epsilon_0 R^3 + \frac{e^2}{\omega_0^2}\left[\frac{1}{m} + \frac{1}{M}\right] + \frac{\mu^2}{3KT}\right\}$$

Thus, the equation is known as the Langevin - Debye equation

5) If a NaCl crystal subjected to an electric f
of 1000 V/m and the resulting polarisation is
$4.3 \times 10^{8} C/m^2$, calculate the dielectric con
of NaCl.

$\Rightarrow$ $E = 1000 V/m$

$P = 4.3 \times 10^{8} \, c/m^2$

$\epsilon_0 = 8.85 \times 10^{-12}$

$P = \epsilon_0 (\epsilon_r - 1) E$

$\frac{P}{E} = \epsilon_0 (\epsilon_r - 1)$

$\frac{P}{E\epsilon_0} + 1 = \epsilon_r$

$\epsilon_r = \frac{4.3 \times 10^{8}}{1000 \times 8.85 \times 10^{-12}} + 1$

$\epsilon_r = \frac{4.3 \times 10^{-8}}{8.85 \times 10^{-9}}$

$= 4.86 + 1$

$\epsilon_r = \boxed{5.86}$